

Storage warehouse

Revise 360 lesson plan — 60 minutes

Unit	OCR J277 1.2 Memory and storage — Paper 1, Computer systems
Lesson	Storage warehouse (lesson 3 of the unit)
Lesson ID	ms-l03
Experience ID	ms-l03
Plan ID	PLAN-MS-L03
Content version	dbdd69566dfa
Website	https://revise360.co.uk/experience.html?id=ms-l03

Big question, outcomes and success criteria

Big question — Why do computers have secondary storage?

Prerequisites

The lesson before this one is ms-l02. Pupils should already have finished it: the starter below retrieves from it.

Earlier in the unit: ms-l01.

Assessable outcomes

1. **ms-l03-o1** — understand the need for secondary storage
2. **ms-l03-o2** — know the common types of storage
3. **ms-l03-o3** — know how each type works

Vocabulary, resources and preparation

Key vocabulary — Secondary storage, Non-volatile

Resources

SLIDES	presentations/MS_L03_StorageWarehouse_Lesson.pptx
WORKSHEET	worksheets/MS_L03_StorageWarehouse_Worksheet.docx
ONLINE EXPERIENCE	experience.html?id=ms-l03

Before the lesson

1. Sign in on the **teacher dashboard** (teacher.html) and open **Class logins**. Enter the class or group and the usernames, one per line, then print the login cards. A pupil cannot sign in without one, so do this before the lesson rather than in it.
2. Print the worksheet, one per pupil.
3. Open the slides and the experience once yourself, on the machine you will be teaching from, so the 360 image is already cached.
4. Check headsets or devices are charged and on the school network.

How pupils get in

1. Go to `topics.html` on revise360.co.uk.
2. Sign in with the **username and PIN from the login card**. The class is already set against the card.
3. Choose the topic, then this lesson's card.

Worth knowing before you promise anything. In this build `APP_CONFIG.backendUrl` is empty, so logins, login cards and progress are stored on the device the pupil used. The teacher dashboard says so too. A pupil who moves to a different machine starts again, so keep a class on the same devices within a unit, and treat the paper worksheet as the record that travels.

Teaching sequence

Sixty minutes including transitions. The online block is the one to move if your lesson is shorter: see the adaptation at the end.

MIN	PHASE	TEACHER	PUPILS	EVIDENCE TO CHECK
0-5	Retrieve	Starter on the board or the worksheet, from the last lesson and this lesson's prerequisites.	Answer the starter on the worksheet.	The starter box on MS_L03_StorageWarehouse_Worksheet.docx is filled in.
5-8	Review the starter	Take answers; correct the one most got wrong.	Correct their own starter.	A correction written, not just a tick.
8-18	Explain and model	The new idea in two or three steps, with the example worked in front of the class.	Listen and answer questions about the example.	They can say why, not only what.
18-21	Set up and settle	Sign-in, launch, and what to write before anyone taps an answer.	Sign in, open the lesson, find Why secondary storage?.	Everyone is in and on station 1.
21-43	Online: work the stations in order	Pupils read each wall, write the key fact and the challenge, then answer at the badge.	Read each wall, write the key fact and the challenge, then answer at the badge.	The worksheet has writing on it for each of the 6 stations before the badge is tapped.
43-48	Check understanding	The hinge question, hands down.	Answer the hinge question, hands down.	The split across the options.
48-56	Apply	Exam practice on the worksheet, in silence.	Exam practice, in silence.	worksheet, Exam practice a) to c)
56-60	Exit and set up the next lesson	The exit question and the retrieval link to next lesson.	Write the exit answer.	One sentence that answers the big question.

The explanation and the worked example

This is what the walls of the experience say, station by station. Teach the first two or three from the front before anyone puts a headset on; the rest the pupils read for themselves.

1. Why secondary storage?

- It is non-volatile, so data is kept when the power is off.
- It holds programs and files that aren't currently in use.
- It is much larger, and much cheaper per gigabyte, than RAM.

Challenge on this wall: Why do computers have secondary storage as well as RAM?

2. Magnetic storage

- Hard disk drives use spinning platters coated in magnetic material.
- A read/write head moves across the surface, magnetising tiny areas as 1s and 0s.
- Magnetic tape works the same way, but reads in sequence from a long strip.

Challenge on this wall: Why are moving parts a problem for a laptop's storage?

3. Optical storage

- CDs, DVDs and Blu-ray discs are read by a laser.
- Data is stored as pits and lands burned into the surface, which reflect the laser differently.
- Discs are cheap, light and portable, but hold relatively little and scratch easily.

Challenge on this wall: Rank CD, DVD and Blu-ray by capacity.

4. Solid state storage

- SSDs, USB sticks and memory cards use flash memory.
- Data is stored as electrical charge in cells, with no moving parts.
- Very fast, silent, durable and light, but more expensive per gigabyte.

Challenge on this wall: Why is solid state best for a laptop that gets carried around?

5. Inside the devices

- Magnetic: platters, spindle, read/write head, actuator arm.
- Optical: laser, lens, spinning disc, pits and lands.
- Solid state: controller chip and flash memory chips.

Challenge on this wall: Label the parts of a hard disk on your worksheet.

6. Choosing a type

- Magnetic: huge capacity, cheap, but slower and fragile when moved.
- Optical: cheap and portable, small capacity, easily scratched.
- Solid state: fast and durable, but dearer per gigabyte.

Challenge on this wall: Which type would you fit in a school's file server, and why?

Retrieval and the check for understanding

Starter — List everything stored on the computer you are using that must still be there tomorrow morning.

Answer space: 3 lines on the worksheet.

The answer is in the teacher answers for this unit, not on this plan: a plan left on a desk should not give it away.

Hinge question

Use this one, hands down, before anyone moves on to independent work. It is a real activity from the experience (ms-l03-s3-c1), so the wording a pupil sees matches the wording you say.

How is data read from an optical disc?

Correct: A laser detects pits and lands on the surface

What a pupil might choose instead

- A magnet reads the platters
- Electricity is trapped in flash cells
- A needle follows a groove

Count the hands on each. If more than a few of the class are on one wrong option, that option is the misconception to teach against before anyone goes on: go back to the station this question sits on and work through its example again, aloud, before the independent work.

The course's own response to a wrong answer here: Blu-ray uses a blue laser with a shorter wavelength, so it fits more data.

One response, not four. one response covers every wrong option; it does not say what to revisit for each one, so decide your own reteach for each option before the lesson - the list above is the start of it.

Guided to independent practice

STAGE	ACTIVITIES	WHAT IT IS
Recognise	10	The answer is on the screen and the pupil picks it: multiple choice, choosing several, matching a pair.
Construct	2	Nothing is offered. The pupil classifies, orders, fills in a table or works a number out, and types or places the answer.

Scaffolds already in the product, and how they fade

- **Hint** is a ladder, not a door: the first rung names the idea, the next shows the shape of the code, the next how it starts, the last animates the technique on different data. A pupil asks for each rung. None of them is the answer.
- **Run** is free and unmarked; **Check my answer** marks. A pupil may run as often as they like, which is what makes trial and error safe.
- **I need help** says to ask you, and says plainly that the question still has to be completed. It does not notify anyone and it does not unlock anything.
- The worked example appears on every early activity and on none of the late ones. That fade is deliberate; it is not a missing example.

Accessibility

- **Read aloud** on every task window reads the task and its brief, with the prescribed values read as values rather than as punctuation.
- Text is 16px or larger throughout, and a question window scrolls rather than shrinking its type.
- Prescribed data is boxed and in a monospace face as well as coloured, so it still reads as code in black and white, in a high-contrast mode, and to a pupil who does not separate green from amber.
- **Lower literacy and EAL:** pre-teach the vocabulary above; let the pupil say the key fact before writing it; accept the key fact in two or three words.
- **SEND:** the sequence is compulsory but the pace is not. A pupil may finish a station next lesson; the progress saves after every answer and reopens at their earliest unfinished question.
- A headset is never required. Every experience works on a laptop or a tablet with a mouse or a finger.

Delivering the online experience

Before anyone enters

- Worksheet out, pen in hand. The rule is write first, then answer at the badge: the first answer is the one that counts.
- Say where the stations are. They are numbered around the room: 1 and 2 to the right, 3 and 4 behind, 5 and 6 to the left, and the final challenge on the floor.
- Say that a badge marked **Not yet** is not a fault. It opens when the station they are on is finished.

The stations, and what to record at each

#	STATION	ACTIVITIES	RECORD ON PAPER
1	Why secondary storage?	2	Secondary storage is needed because...
2	Magnetic storage	2	Magnetic storage works by...
3	Optical storage	2	Optical storage works by...
4	Solid state storage	2	Solid state storage works by...
5	Inside the devices	1	The main parts are...
6	Choosing a type	2	Each type suits...
final	Final challenge: sort the storage	1	the answer to the station's challenge

Where to pause

Stop the class after station 2 and take answers to its challenge out loud. That is the point where a pupil who has misread the technique can still be put right cheaply.

Afterwards

Headsets off and down. Ask three pupils for the key fact from a station of your choosing, in their own words and not read off the sheet. Then the exit question.

Assessment, evidence and unfinished work

ID	QUESTION	MARKS	WHERE
ms-l03-x1	State what is meant by non-volatile storage.	1	worksheet, Exam practice a); deck, Exam practice slide

ID	QUESTION	MARKS	WHERE
ms - 103 - x2	Describe how a hard disk drive stores data.	2	worksheet, Exam practice b); deck, Exam practice slide
ms - 103 - x3	Describe how data is read from an optical disc.	2	worksheet, Exam practice c); deck, Exam practice slide

Expected answers

Model answers and marking guidance are teacher-only and are not printed on this plan. They are in the unit's teacher guide pack, and the per-activity marking for this lesson is in `answers/records/ms-103.json`, which is not published with the website.

Common errors

The wrong options the activities offer are the errors this lesson expects. The commonest to watch for:

- **Why do computers need secondary storage?** — Because the CPU works from it directly; Because it is faster than RAM
- **Which statement is true?** — Secondary storage is faster than RAM; Secondary storage is volatile
- **How does a hard disk drive store data?** — By burning pits with a laser; In flash memory cells with no moving parts
- **What is magnetic tape mostly used for now?** — Running the operating system; Storing photos on a phone
- **How is data read from an optical disc?** — A magnet reads the platters; Electricity is trapped in flash cells

Unfinished work

Record where each pupil stopped, not a tick. They resume at their earliest unfinished question, and a station left half-done is the thing to open next lesson. Do not mark the lesson complete for a pupil who did not finish it: the score screen already shows what was and was not done, and that is the honest record.

Exit, next lesson and homework

Exit check

One of these, written, not a show of hands:

- The need for secondary storage
- Magnetic, optical and solid state
- How each type works

The confidence columns on the worksheet are a supplement to this, not a substitute: a pupil who rates themselves confident and cannot answer the exit question has told you nothing.

Retrieval link to next lesson

Next is `ms - 104`, Top Trumps. Use this lesson's big question as next lesson's starter, so the first thing they do is retrieve it.

Homework

Finish any station left open, on the website, from the question they stopped at. If the class has no devices at home, set the exam practice on the worksheet instead and mark it next lesson.

Shorter lessons, longer lessons and no internet

A 40 to 45 minute lesson

Keep the starter, the explanation and the model at full length: cutting those is what makes the online work fail. Take the time out of the online block and the apply phase. Pupils stop where they are; their progress saves after every answer and reopens at their earliest unfinished question, so the next lesson starts there.

If the internet or the devices are unavailable

Teach from the slides and the worksheet. The slides carry the same teaching as the station walls, and the worksheet carries the key facts, the challenges and the exam practice. Set the online stations when the devices are back.

There is no offline version of the experience, and this plan does not contain one. Do not hand out a printed walkthrough of the stations as a substitute: the experiences are used on the website.